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Key Points:

- A new crustal shear-wave velocity model beneath the Pamir–Tien Shan is constructed utilizing adjoint-state traveltime tomography
- Major active faults in the Pamir interior are rooted in three deep-crustal low-velocity zones containing low-degree partial melts
- Melt-induced rock weakening facilitates fault localization and modulates modern crustal kinematics of the Pamir

Supporting Information:

Supporting Information may be found in the online version of this article.

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Melt-Regulated Crustal Kinematics in the Pamir, NW Tibetan Plateau: Insights From Surface-Wave Tomography

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Abstract The Pamir, located at the northwestern margin of the Tibetan Plateau, is undergoing E–W extension within its interior, accommodated by micro-block motions along major fault systems; however, the mechanisms governing this modern block kinematics remain elusive. Here, we present an updated crustal shear-wave velocity model of the region, developed via adjoint-state surface-wave traveltime tomography. The Rayleigh-wave dispersion data set is compiled from teleseismic and regional earthquakes, as well as ambient noise; for the latter, three-station interferometry was employed to incorporate asynchronous records. The resulting model reveals distinct middle-to-lower crustal low-velocity zones (LVZs) that spatially align with the major active faults bounding the micro-blocks. Although the estimated melt fractions within these LVZs fall below the rheological threshold required to sustain large-scale flow, rock strength is drastically reduced within these zones. This melt-induced weakening focuses deep-seated strain and promotes fault development, thereby regulating the modern crustal kinematics of the Pamir.

Plain Language Summary Analogous to the interior of the Tibetan Plateau, the Pamir at the northwestern margin of Tibet is undergoing E–W extension, during which the crust breaks apart and slips along major faults that frequently generate damaging earthquakes. Yet the factors controlling the development of these faults remain poorly understood. In this study, we exploit seismic-wave traveltime data recorded by 538 seismograph stations to construct a high-resolution, three-dimensional model of crustal seismic velocity beneath the region. Our results show that major surface faults are rooted in discrete deep-crustal zones containing small amounts of melt. Although the melt volume is insufficient to enable rapid, large-scale mass flow, it substantially weakens the surrounding rocks. This weakening promotes the absorption of tectonic stress and determines where the crust fractures. These findings demonstrate that small amounts of melt can act as a key regulator of fault positioning, providing new insights into the mountain-building processes of orogenic systems.

1. Introduction

The Pamir, located at the northwestern margin of the Tibetan Plateau, accommodates ~40% of the total India–Asia convergence (Wang and Shen, 2020) and exhibits complex deformation styles during its Cenozoic structural evolution. Geodetic observations (Ischuk et al., 2013; Wang et al., 2024; Zhou et al., 2016) show that the arcuate Pamir is bounded by several major fault systems (Figures 1b and 2e): to the north, the dextral-reverse Pamir Thrust System (PTS), separating it from the Tien Shan, accommodates ~7–17 mm/yr shortening and ~9–11 mm/yr dextral shear; to the east, the Kashgar–Yecheng transfer system (KYTS) separates the Pamir from the Tarim Basin with ~2–4 mm/yr shortening and ~2.7 mm/yr dextral shear; and to the west, the Darvaz Fault (DF) bounds the Tajik Basin, accommodating ~4–10 mm/yr sinistral shear and possibly ~4–9 mm/yr shortening. These boundary-defining faults generally initiated between 25 and 18 Ma, with activity levels varying over time (Coutand et al., 2002; Sobel and Dumitru, 1997). Contemporaneously, deformation within the Pamir interior was dominated by orogen-normal extension that exhumed the Yazgulom–Sarez–Muskol–Shatput and Shakh-dara–Alichur domes parallel to Mesozoic sutures (Rutte et al., 2017; Worthington et al., 2020). However, this internal deformation style has since evolved. Modern kinematics, as suggested by elastic block modeling (Wang et al., 2024), do not exploit these ancestral Mesozoic weaknesses; instead, they are characterized by the relative shearing and divergence of three micro-blocks along the nascent, NNE-trending sinistral Sarez–Karakul Fault System (SKFS) and the SSE-trending dextral Kongur Shan Extensional System (KSES) (Figure 1b). This configuration results in an overall counterclockwise rotation of the Pamir, accompanied by internal E–W

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extension. Several destructive earthquakes have occurred along these active fault systems in recent years, including the 2023 Mw 6.8 Murghab, 2016 Mw 6.6 Muji, and 2015 Mw 7.2 Sarez earthquakes (Figure 1b).

Despite these well-documented block kinematics, the mechanisms controlling modern fault development in the Pamir remain poorly constrained compared to the main collisional belt of Tibet (Copley et al., 2011; Nie et al., 2023; Zhang et al., 2023). This issue is particularly relevant given our recent findings in southern Tibet that intersections between neotectonic faults and preexisting asperity-like gneiss belts may be susceptible to large earthquakes (Zhang et al., 2025), a scenario likely applicable to the 2015 Sarez earthquake (Sangha et al., 2017) (Figure 1b). The SKFS has been interpreted as a transfer corridor that mirrors the lithospheric boundary of the underthrusting Indian slab at mantle depths (Kufner et al., 2018), whereas the KSES has been attributed to near-surface kinematic processes, such as radial thrusting or northward propagation of the Karakoram Fault (Ge et al., 2022). Nevertheless, the relationship between faulting and crustal architecture remains elusive, despite the key role of crustal processes in governing the mechanical weaknesses and regional stress fields that ultimately control fault localization and kinematics.

Geophysical investigations provide a critical window into these processes, establishing a basic tectonic framework for the Pamir and Hindu Kush that involves north-propagating in situ lithospheric delamination beneath the Pamir and north-directed Indian marginal slab subduction and break-off beneath the Hindu Kush (Gao et al., 2025; Kufner et al., 2021; Li et al., 2018; Pan et al., 2026). Both processes are marked by intermediate-depth seismicity belts (Sippl, Schurr, Yuan, et al., 2013) (blue squares in Figure 1b). The crust beneath the Pamir is exceptionally thick, averaging ~65–75 km and locally reaching 80–90 km (Jia et al., 2026; Schneider et al., 2019; Xu et al., 2021) (Figure S1 in Supporting Information S1). Low-Vs anomalies pervasively characterize the deep crust, although the delineated geometries vary in different studies (Gao et al., 2025; Kumar et al., 2022; Li et al., 2018; Liang et al., 2020). These anomalies are indicative of the thick crust and have also been interpreted as evidence of partial melting or even viscous flow; however, the degree of melting remains unclear.

In this study, to gain insights into the deep-seated control over the Pamir's characteristic kinematics, we develop a new shear-wave velocity model of the region at significantly improved resolution, enabled by a greatly expanded surface-wave data set and state-of-the-art seismic imaging techniques. Our model reveals crustal melt zones linked to active faults, highlighting the key role of melt in regulating fault development and crustal kinematics within the Pamir Plateau.

2. Rayleigh-Wave Dispersion Data

To maximize the resolution of lithospheric structures beneath the Pamir and Tien Shan, we compiled a comprehensive Rayleigh-wave dispersion data set using vertical-component seismograms from both regional and global seismic networks within a 45° great-circle radius of study region ($n = 538$; Table S1 in Supporting Information S1 and Figures 1c and 1d). The data set integrates phase and group velocity measurements derived from ambient noise, teleseismic waveforms, and regional earthquakes.

Ambient noise phase velocity. Two-station interferograms (C_2) were first retrieved between synchronously operating stations following Bensen et al. (2007). Direct-wave three-station interferometry (Zhang et al., 2020) was subsequently applied to the C_2 interferograms. Given the availability of C_2 between a common virtual source-station and two receiver-stations, three-station interferograms (C_3) between receiver-stations can be constructed by correlating or convolving the Rayleigh-wave segments of the corresponding C_2 (Figure S2d in Supporting Information S1). This approach bypasses the requirement for simultaneous station operation by utilizing long-running stations as bridges, thereby significantly increasing the spatial coverage of the dispersion data set. Fundamental-mode phase velocity dispersions (3–55 s) were subsequently extracted from both C_2 and C_3 interferograms using Automatic Frequency–Time Analysis (AFTAN) (Bensen et al., 2007).

Teleseismic earthquake phase velocity. To complement the ambient noise data at longer periods (>15 s), we applied the two-station image analysis (Yao et al., 2005) to teleseismic waveforms. A systematic discrepancy was observed where ambient noise phase velocities were slightly slower (~30 m/s on average) than those derived from teleseismic measurements (Figure S3a in Supporting Information S1). Inversions of the ambient noise source distribution indicate negative source-induced phase velocity biases near NNW–SSE directions (Figures S4a and S4e in Supporting Information S1), where most station pairs are preferentially oriented (Figure S4b in Supporting Information S1) (Hao et al., 2026; Harmon et al., 2010). Thus, biased azimuthal sampling provides a first-order

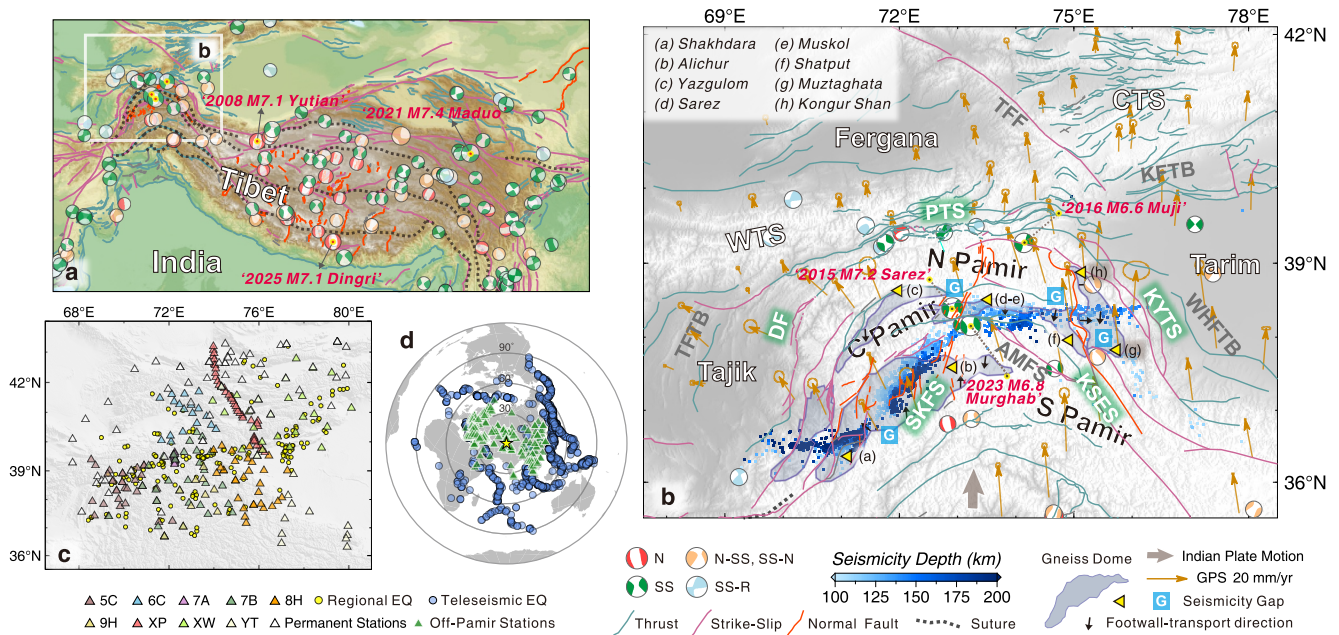


Figure 1. Tectonic setting of the study region. (a) Regional topographic overview showing active fault systems and associated large-magnitude crustal earthquakes within the Tibetan Plateau. Focal mechanism solutions are from the GCMT catalog (2000–2025; $M_w \geq 5.5$). (b) Detailed map of the Pamir (NW Tibet). Faults from the Central Asia Fault Database (Mohadjer et al., 2016) are color-coded by type, with major systems highlighted in green. Yellow triangles mark Neogene gneiss domes. Black arrows show the dominant kinematics (footwall–transport directions) of dome-bounding shear zones (Worthington et al., 2020), orange arrows show GPS velocities (Wang et al., 2024), and the thick gray arrow shows Indian Plate motion (DeMets et al., 2010). Squares show intermediate-depth seismicity colored by depth (Bloch et al., 2021; Sippl, Schurr, Yuan, et al., 2013), with seismicity gaps (G) annotated. (c) Regional earthquakes and seismic stations utilized in this study. (d) Teleseismic earthquakes and off-study-region seismic stations (virtual sources) utilized in this study. Abbreviations: DF, Darvaz Fault; PTS, Pamir Thrust System; KYTS, Kashgar–Yecheng Transfer System; SKFS, Sarez–Karakul Fault System; KSES, Kongur Shan Extensional System; AMFS, Aksu–Murgab Fault System; TFTB, Tajik Fold–Thrust Belt; WHFTB, Wupaer–Hotan Fold–Thrust Belt; KFTB, Kepingtage Thrust–Fold Belt; TFF, Talas–Ferghana Fault; WTS, Western Tien Shan; CTS, Central Tien Shan.

explanation for the observed discrepancy. Nevertheless, the observed and predicted bias patterns still differ in both detail and magnitude. The remaining discrepancies may also arise from departures from the far field approximation and propagation effects associated with structural heterogeneity, anisotropy, and scattering (Harmon et al., 2010; Yao and van der Hilst, 2009). To ensure inter-dataset compatibility, we calibrated the ambient-noise-derived phase velocities using empirical, period- and interstation-distance-dependent average differences between the two data sets (Figure S3c in Supporting Information S1) (Timkó et al., 2024).

Regional earthquake group velocity. Finally, we utilized DisperPy (Nascimento et al., 2025), a convolutional neural-network-based picker, to measure Rayleigh-wave group velocity dispersions from regional earthquakes. These measurements provide independent spatial sampling that complements the interstation phase velocity observations. Detailed technical specifications, statistical summaries, and spatial coverage of the three data sets are provided in Text S1, Figures S2, S3, S5, and S6 in Supporting Information S1, respectively.

3. Adjoint-State Surface-Wave Traveltime Tomography

We employ the adjoint-state surface-wave traveltime tomography method SurfATT (Hao et al., 2024; Xu et al., 2025), a direct surface-wave inversion technique with 3-D model parameterization, to invert for shear-wave velocity $\beta(\mathbf{x})$ (i.e., model vector $\mathbf{m}$). The objective of this method is to minimize the weighted frequency-dependent surface-wave traveltime misfit between observations $T_{f,i,j}^o$ and model predictions $T_{f,i,j}(\mathbf{m})$ for all frequencies ($f = 1, 2, \dots, N_f$) and all station pairs ($i, j = 1, 2, \dots, N_r$):

$$\min_{\mathbf{m}} \chi(\mathbf{m}) = \sum_{f=1}^{N_f} \chi_f(\mathbf{m}) = \sum_{f=1}^{N_f} \sum_{i=1}^{N_r} \sum_{j=1}^{N_r} \frac{w_{f,i,j}}{2} (T_{f,i,j}(\mathbf{m}) - T_{f,i,j}^o)^2.$$

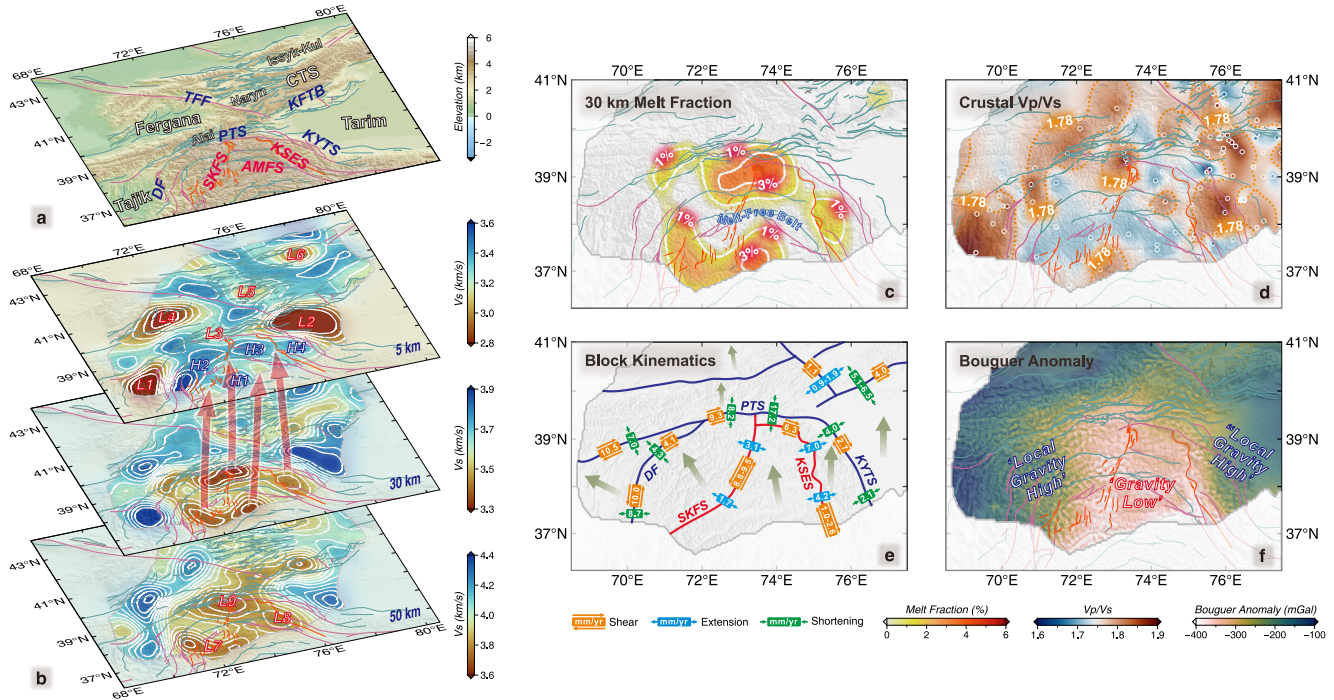


Figure 2. Seismic tomographic images of the crust beneath the Pamir–Tien Shan and surrounding regions. (a) Tectonic framework showing major tectonic units and bounding fault systems. (b) Vs displayed as horizontal slices at 5, 30, and 50 km depths. Prominent anomalies are labeled at 5 and 50 km. L and H denote low- and high-velocity anomalies, respectively. Inferred melt pathways beneath Pamir fault systems are indicated by red arrows. The continuity of crustal low-Vs corridors is further illustrated by 5-km-interval horizontal Vs slices in Figure S10 in Supporting Information S1. Poorly resolved regions are masked in all panels. (c) Melt fractions at 30 km depth, focused on the Pamir. (d) Average crustal Vp/Vs ratio, constrained by receiver functions (Schneider et al., 2019; Zhang and Bao, 2023). Color-filled circles denote measurement sites. (e) Present-day block kinematics constrained by GPS velocities (Wang et al., 2024); dark green arrows indicate principal micro-block motions relative to stable Eurasia. (f) Bouguer gravity anomaly (Bonvalot et al., 2012). The pronounced negative anomaly in the East Pamir generates a significant E–W gradient in GPE.

The Fréchet derivative $K_c^{\beta;\alpha,\rho}(\mathbf{x})$ of the misfit function with respect to the 3-D shear-wave velocity $\beta(\mathbf{x})$ is calculated by combining the 2-D surface-wave kernel $K_f^s(\mathbf{x}^0)$ with the 1-D depth sensitivity kernels of surface-wave velocity c with respect to Vp, Vs, and density, namely, $K_{c,f}^\alpha(\mathbf{x})$, $K_{c,f}^\beta(\mathbf{x})$, and $K_{c,f}^\rho(\mathbf{x})$:

$$K_c^{\beta;\alpha,\rho}(\mathbf{x}) = \left\{ \sum_{f=1}^{N_f} -K_f^s(\mathbf{x}^0) s_f(\mathbf{x}^0) \left[K_{c,f}^\alpha(\mathbf{x}) \frac{d\alpha(\mathbf{x})}{d\beta(\mathbf{x})} + K_{c,f}^\beta(\mathbf{x}) + K_{c,f}^\rho(\mathbf{x}) \frac{d\rho(\mathbf{x})}{d\beta(\mathbf{x})} \right] \right\} \beta(\mathbf{x}).$$

Here $s_f(\mathbf{x}^0)$ is the surface-wave slowness derived from $\beta(\mathbf{x})$ under the assumption of a horizontally homogeneous model. The terms $d\alpha(\mathbf{x})/d\beta(\mathbf{x})$ and $d\rho(\mathbf{x})/d\beta(\mathbf{x})$ represent empirical relationships linking Vp and density to Vs. Surface-wave traveltime fields and kernels are computed with high accuracy and efficiency by solving the eikonal and adjoint equations using the fast-sweeping method (Zhao, 2004); whereas the depth kernels are calculated using sreg96 (Herrmann, 2013) based on the variational principle. Model updating is handled by a line search scheme, with the search direction determined by the L–BFGS algorithm. Detailed technical specifications of the SurfATT setup are provided in Text S2 in Supporting Information S1.

4. Results

Details of the checkerboard and reliability tests used to assess the resolvability and robustness of the final tomographic model are provided in Texts S3 and S4 in Supporting Information S1, respectively. The checkerboard tests indicate that structural features with characteristic lateral dimensions of $\sim 1.0^\circ \times 1.0^\circ$ can be well resolved above ~ 60 km depth, with vertical resolution of ~ 10 km in the upper crust and ~ 20 – 30 km in the deep crust (Figures S7 and S8 in Supporting Information S1). The velocity anomalies discussed in this study are comparable to or larger than these resolution limits.

In the shallow crust ($< \sim 17$ km), low- V_s anomalies are primarily observed beneath the major sedimentary basins (Figure 2b and Figure S9 in Supporting Information S1), including the Tajik, Tarim, and Fergana basins (L1, L2, L4), as well as intermontane basins such as the Alai, Naryn, and Issyk-Kul basins (L3, L5, L6). In particular, V_s decreases to below 2.8 km/s at ~ 5 km depth beneath the major basins. In contrast, relatively high V_s values characterize much of the Pamir and Tien Shan mountain ranges. At greater depths, V_s is generally lower beneath the mountain ranges, where the crust is thicker, and higher beneath the adjacent basins, where the Moho is shallower (Figure 2b and Figure S1 in Supporting Information S1).

Beyond these expected regional features, our model unveils fine-scale internal structures within the Pamir crust (Figure 2b). The high-velocity shallow crust of the Pamir is subdivided into four patches (H1–H4), bounded by active fault systems: the normal-sinistral SKFS, the normal-dextral KSES, and the dextral Aksu-Murgab Fault System (AMFS). The reduced velocities associated with these fault systems can be traced downward into the middle-to-lower crust, where they merge with three distinct low-velocity zones (L7–L9). These connections appear subvertical beneath the SKFS and KSES, which define the micro-block divisions (Wang et al., 2024), while the low-velocity signature associated with the subordinate AMFS dips southwestward, eventually converging with L7 beneath SKFS at depth. A more detailed visualization of these trans-crustal low-velocity channels is provided in Figure S10 in Supporting Information S1, which displays horizontal slices from 5 to 55 km at 5-km intervals.

5. Discussion

5.1. Comparison With Previous Tomography Studies

Our updated crustal V_s model (Figure 2b), established from a significantly expanded surface-wave data set, is in agreement with previous tomographic studies regarding the large-scale structural heterogeneities between basins and ranges (Gao et al., 2025; Kumar et al., 2022; Li et al., 2018; Lü et al., 2019; Sippl, Schurr, Tympel, et al., 2013; Sun et al., 2022) (Figure S11 in Supporting Information S1). However, distinct discrepancies exist regarding the internal features of the Pamir.

Previous surface-wave-based V_s models suggested a blurred spatial correlation between deep-crustal LVZs and the Pamir gneiss domes. Although consensus exists regarding the Shakh-dara–Alichur domes (Kumar et al., 2022; Li et al., 2018; Liang et al., 2020), discrepancies persist for the Yazgulom–Sarez–Muskol–Shatput (Li et al., 2018) and Muztaghata–Kongur Shan domes (Kumar et al., 2022). Our model broadly concurs with the delineation by Kumar et al. (2022) (Figure S11 in Supporting Information S1). In contrast, local V_p model reveals a prominent E–W velocity contrast in the deep crust, with higher velocities in the west and lower velocities in the east (lowest south of the intermediate-depth seismicity belt) (Sippl, Schurr, Tympel, et al., 2013). The discrepancy between V_s and V_p structures likely reflects their differing sensitivities (Watanabe, 1993): V_s is more responsive to partial melting, whereas V_p can be more strongly influenced by the thermal state. In this context, the V_p contrast is consistent with a relatively colder region west of the inferred delaminating Pamir lithosphere and a comparatively hotter regime to the east. The delamination scenario is based on previous mantle-depth observations, including the intermediate-depth seismicity belt (Sippl, Schurr, Yuan, et al., 2013) (Figure 1b) and slab-like high-velocity mantle anomalies (Kufner et al., 2016). Moreover, in the shallow crust, where previous models showed weak geology–velocity correlations and limited inter-model consistency (e.g., Li et al., 2018; Sippl, Schurr, Tympel, et al., 2013), our results resolve a clear micro-block division bounded by active fault systems that connect downward to deep-crustal LVZs.

5.2. Low-Degree Partial Melting in the Pamir Deep Crust

The delineation of L7–L9, characterized by V_s values of < 3.45 km/s, < 3.5 km/s, and < 3.4 km/s at 30 km depth, respectively, indicates the presence of partial melting within the Pamir crust. Using Gassmann's equation (Chu et al., 2010) and a 30-km-depth horizontal slice for illustration, we quantified melt fractions in the middle crust beneath the mountainous regions. The results reveal pervasive low-degree melting across the Pamir Plateau, with the exception of a $\sim$ WNW–ESE-trending melt-free belt (Figure 2c and Figure S12 in Supporting Information S1). Estimated melt fractions within L7–L9 beneath the SKFS and KSES range from $\sim 1\%$ to $\sim 4\%$, whereas melting is negligible beneath the Tien Shan.

The melt fraction calculation uses 3.54 km/s as the reference velocity for the unmolten solid framework. This value corresponds to the maximum of the Pamir regional-average 1-D Vs profile within the 0–35 km depth range, which occurs at ~12 km depth (Figure S13a in Supporting Information S1). Uncertainties were evaluated by adopting a broad velocity range of 3.48–3.73 km/s for the solid frame, encompassing representative middle-crustal lithologies over a temperature range of 467–786°C (Christensen & Mooney, 1995), alongside a wide range of secondary parameters (e.g., critical porosity of 30–35 vol.% and water saturation of 0.0–0.5 wt.%). Detailed formulas and uncertainty analyses are provided in Text S5 in Supporting Information S1. The results indicate that even under extreme parameter combinations (Figures S12g and S12i in Supporting Information S1), partial melting persists in the Pamir Plateau, with most regions characterized by low melt fractions (<7%). Further comparison with the radial anisotropy model of Liang et al. (2020) (Figure S11d in Supporting Information S1) suggests that positive radial anisotropy around L7 and L9 may partly account for their higher estimated melt fractions relative to L8. Nevertheless, the inference of low-degree crustal partial melting beneath the major fault systems of the Pamir remains robust.

The inferred spatial distribution of crustal melting in the Pamir is corroborated by multiple independent geophysical and geological observations. (a) Vp/Vs ratios: The melt distribution closely resembles crustal Vp/Vs patterns derived from receiver functions (Schneider et al., 2019), which show a correspondence between high Vp/Vs ratios (>1.78) and melt-bearing regions, and lower ratios in melt-free areas (Figure 2d). (b) Electrical resistivity: Magnetotelluric data along a ~NNW–SSE profile at ~73.5°E reveal low electrical resistivity (~1 Ωm) coincident with the L7 and L9 anomalies, contrasted by high resistivity (~10³ Ωm) in the intervening melt-free zone (Sass et al., 2014). (c) Geothermal activity: Elevated surface heat flow (>120 mW/m²) and the distribution of hot springs spatially coincide with the L7–L8 anomalies (Chen et al., 2025; Duchkov et al., 2001). (d) Petrology: Migmatite outcrops, products of partial melting in metapelites, occur in the Shakh-dara and Muztaghata domes situated above L7 and L8 (Stübner et al., 2013; Xu et al., 2023), whereas they are rare in the Yazgulom–Sarez–Muskol–Shatput domes located above melt-free regions (Figure 1b).

The melts may partially derive from mantle magmatic upwelling, which likely intensifies proximal to gaps in and around the delaminating Pamir lithosphere, as marked by reduced intermediate-depth seismicity (Bloch et al., 2021; Sippl, Schurr, Yuan, et al., 2013) (labeled “G” in Figure 1b). Additionally, they may originate from in situ crustal melting facilitated by fluids released during dehydration reactions within the delaminating lithosphere (W Li et al., 2020; Sass et al., 2014). This fluid transport lowers the solidus of the host crustal rocks, providing a viable mechanism for the coexistence of relatively high Vp (Gao et al., 2025; Sippl, Schurr, Tympel, et al., 2013); consistent with lower temperatures) and low Vs (this study; indicative of partial melt) beneath the western Pamir.

Furthermore, the Yazgulom–Sarez–Muskol–Shatput domes exhibit “top-borderland” (top-N) kinematics, which is fundamentally distinct from the “top-hinterland” kinematics displayed by the Shakh-dara–Alichur (top-SSE) and Muztaghata–Kongur Shan (top-W) domes (Worthington et al., 2020) (black arrows in Figure 1b). When integrated with our mapped melt distribution and the petrological evidence outlined above, these observations provide a dual framework for Pamir dome formation. The Yazgulom–Sarez–Muskol–Shatput domes originate from the passive deformation by gravitational sliding along the former plateau margin, whereas the Shakh-dara–Alichur and Muztaghata–Kongur Shan domes record the proactive ascent of crustal material facilitated by elevated melt fractions.

5.3. Controlling Factors of Fault Development and Block Kinematics

Our observations demonstrate that active fault systems within the Pamir Plateau, including the major structures governing its kinematic framework, are downward connected to zones of low-degree crustal melting (Figure 2b). This relationship suggests a deep-seated control over fault development and contemporary block kinematics. A key question, however, is the role played by melt in this process and whether it represents the sole or primary controlling factor.

The estimated melt fractions, which remain below the rheologically critical melt percentage (RCMP) of ~7% (Rosenberg and Handy, 2005), indicate that interconnected melt networks are unlikely to be pervasively developed. Consequently, the classic large-scale fast channel-flow model proposed for Tibet (Clark and Royden, 2000) is unlikely to operate in the Pamir, the northwestern margin of the Tibetan Plateau, under present conditions. This interpretation is consistent with the limited connectivity of LVZs (Figure 2b) and with the moderate extrusion and expansion of Pamir Plateau mass since the early Miocene (~100 km) (Y-P Li et al., 2020),

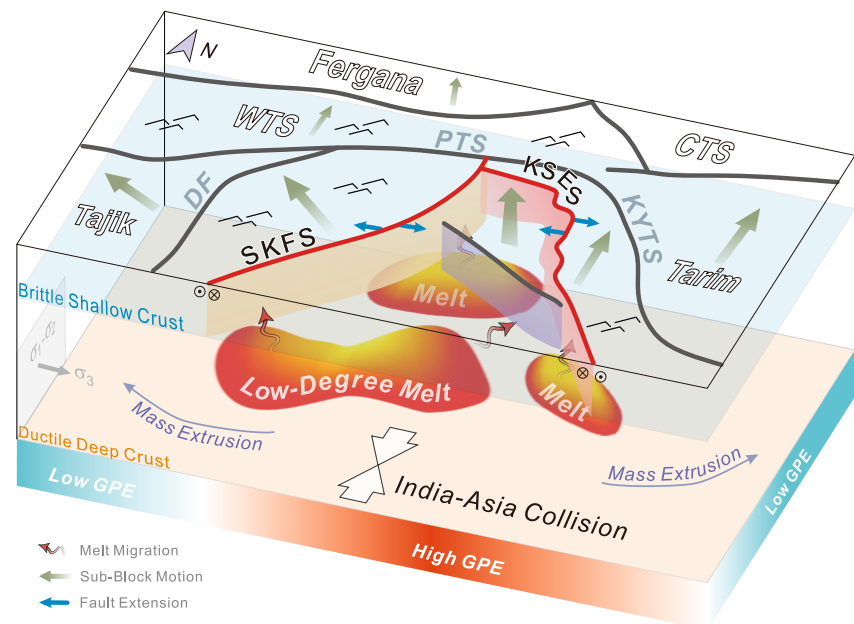


Figure 3. Conceptual cartoons illustrating the controls on fault development and block kinematics in the Pamir, NW Tibet. Micro-block kinematics are modulated by major strike-slip and normal fault systems under an overall E–W extensional regime. The development of these interior faults is facilitated by mechanically weak crustal zones under a favorable regional stress field. In the deep crust, localized low-degree partial melt zones effectively reduce rock strength, thereby regulating the localization of major fault systems. GPE gradients between the high plateau and surrounding lowlands, together with N–S compression from the India–Asia collision, define the regional stress regime, with the minimum principal stress, σ_3 , oriented E–W. Figures are schematic and not to scale.

which aligns with expectations from numerical simulations (Rey et al., 2010). Similar conclusions have been reported for central-eastern Tibet in recent studies (Bao et al., 2020; Hetényi et al., 2011; Li et al., 2024; Nie et al., 2023).

Nevertheless, the most significant reduction in rock strength occurs between the solidus and the RCMP rather than above the RCMP. Laboratory experiments indicate that ~65% of aggregate strength is lost at melt fractions as low as ~4%, owing to the increasing proportion of melt-bearing grain boundaries (Rosenberg and Handy, 2005). Thus, melt fractions comparable to those estimated for L7–L9 may substantially weaken the deep crust, particularly where aqueous fluids released during dehydration reactions further reduce melt viscosity (Rosenberg and Handy, 2005), as is likely beneath the Pamir (W Li et al., 2020). Within the “jelly sandwich” rheological framework (Wang et al., 2021), tectonic strain is expected to localize preferentially within these weakened, partially molten regions. This process promotes the development of ductile shear zones, consistent with field observations (Bürgmann & Dresen, 2008). Where ductile strain weakening is sufficiently rapid, faulting may subsequently initiate near the base of the overlying brittle crust and propagate upward, as suggested by numerical and analogue models (Scholz and Choi, 2022; Sokoutis et al., 2007). Together, these observations establish a mechanically coherent link from melt-bearing LVZs to deep-crustal weakening, strain localization, and fault development (Figure 3).

In this context, melt distribution regulates the spatial positioning of Pamir fault systems and dictates their subdivision into micro-blocks. Similar associations occur elsewhere on the Tibetan Plateau, where several N–S-trending rifts in southern Tibet overlie localized crustal LVZs interpreted to contain low-degree partial melt (Li et al., 2024; Nie et al., 2023). Together with our Pamir observations, these findings suggest a broader role of melt-induced weakening in localizing strain and accommodating contemporary orogen-parallel (~E–W) extension across the high-elevation Tibetan Plateau. Beyond Tibet, melt is also considered critical for localizing deformation and enabling continental breakup in the East African Rift System (Adams et al., 2018; Kendall et al., 2005). Associations between partial melt and fault development also occur in oceanic spreading and subduction-related settings, as exemplified by the Southwest Indian Ridge (Gardner et al., 2020) and the Japanese arc (Nakajima et al., 2006), respectively. Collectively, these observations suggest a recurring role of melt-induced

rheological weakening in strain localization and fault development across diverse tectonic settings. However, this broader perspective does not imply a universal one-to-one correspondence between faults and LVZs, as both can have multiple origins.

Given the overall low melt fractions (Figure 2c) and relative inactivity of the Shakh dara–Alichur domes (Stübner et al., 2013), melt-driven diapirism due to positive buoyancy is likely spatially restricted at present. It may act only as a secondary driver of fault kinematics, for example, along the Kongur Shan segment of the KSES, where active dome exhumation and high extensional rates of $\sim 4\text{--}7$ mm/yr are observed (Cao et al., 2013; Wang et al., 2024). Nevertheless, widespread diapirism under higher melt fractions cannot be ruled out for earlier tectonic episodes. Other tectonic forces must therefore drive the contemporary regional E–W extension accommodated by the dominant SKFS and KSES, which exhibit NNE–SSW and NNW–SSE trends with sinistral-normal and dextral-normal kinematics, respectively. Two principal drivers are considered here (Figure 3):

1. Lateral gradients in Gravitational Potential Energy (GPE). The regional Bouguer gravity pattern (Bonvalot et al., 2012), together with vertically coherent lateral density contrasts throughout the lithospheric column (Yan et al., 2025), supports the existence of a pronounced GPE contrast. This contrast separates the high Pamir interior, particularly the eastern plateau characterized by strongly negative Bouguer gravity anomalies (Figure 2f), from the Tajik and Tarim forelands. The resulting GPE gradient is expected to promote foreland-directed extension toward both the west and east, expressed as ductile extrusion in the deep crust and gravitational sliding in the shallow crust (Rey et al., 2010).
2. N–S axial compression. The northward advancement of the Indian slab and the blockage by the Tien Shan to the north promote E–W directed stretching and extrusion of ductile crustal materials. This provides an additional dragging force upon the brittle shallow crust through viscous buckling (Bischoff & Flesch, 2018). Because the N–S compressional stress is not sufficiently counteracted to rotate the maximum principal stress axis to the vertical, strike-slip motion persists and locally dominates over normal faulting at the SKFS, AMFS and the Muji Fault.

Thermochronological constraints indicate that E–W extension within the Karakul Graben and the Kongur–Muztaghata domes, bounded by the SKFS and KSES, respectively, initiated at $\sim 9\text{--}3$ Ma (Amidon & Hynek, 2010) and $\sim 7\text{--}4$ Ma (Cao et al., 2013; Robinson et al., 2010). Thus, the modern crustal deformation and kinematic style in the Pamir, which superseded the previous margin-normal extension recorded by the Shakh dara–Alichur and Yazgulom–Sarez–Muskol–Shatput domes (Worthington et al., 2020), commenced in the Late Miocene. This tectonic shift broadly coincided with the Pamir–Tien Shan collision (Sun et al., 2015), suggesting that the current configuration of melt, GPE, and stress was fully established during that period.

6. Conclusions

Using an expanded Rayleigh-wave dispersion data set and adjoint-state surface-wave tomography, we present a high-resolution crustal shear-wave velocity model for the Pamir and Tien Shan. The model reveals that the active fault systems governing modern Pamir micro-block kinematics are spatially rooted in discrete deep-crustal LVZs. Estimated melt fractions within these LVZs are low and therefore insufficient to sustain large-scale channel flow or diapirism. However, under the modern stress regime, characterized by E–W extension shaped by GPE gradients and ongoing India–Asia convergence, these melt-bearing zones play a key role in localizing fault systems due to their capacity for substantial rock weakening. Consequently, they regulate the contemporary crustal kinematics in the Pamir.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The Rayleigh-wave dispersion data and the seismic velocity model obtained in this study are provided in the Zenodo repository at <https://doi.org/10.5281/zenodo.20061741> (Zhang, 2026). All raw seismic data are available through the IRIS (<https://ds.iris.edu/ds/nodes/dmc/data/>), GEOFON (<https://geofon.gfz.de/waveform/archive/>), RESIF (<https://ws.resif.fr/>), and ORFEUS (<https://www.orfeus-eu.org/data/>) data centers via the following links and citations: 5C

(<https://doi.org/10.14470/0P7567352807>) (Schurr et al., 2012), 6C (<https://doi.org/10.14470/1P035555>) (Haberland et al., 2009), 7A (<https://doi.org/10.14470/2N7564774946>) (Yuan et al., 2019), 7B (<https://doi.org/10.14470/2O097102>) (Yuan et al., 2008), 8H (<https://doi.org/10.14470/3U7560589977>) (Yuan et al., 2017), 9H (<https://doi.org/10.14470/4U7561589984>) (Schurr et al., 2017), BS (<https://doi.org/10.7914/SN/BS>) (National Institute of Geophysics, Geodesy and Geography, BAS, 1980), CB (<https://doi.org/10.7914/SN/CB>) (Institute of Geophysics China Earthquake Administration, 2000), CD (<https://doi.org/10.7914/SN/CD>) (Albuquerque Seismological Laboratory, USGS, 1986), CK (<https://www.fdsn.org/networks/detail/CK/>), CZ (<https://doi.org/10.7914/SN/CZ>) (Charles University in Prague et al., 1973), G (<https://doi.org/10.18715/GEOSCOPE.G>) (Institut de physique du globe de Paris and École et Observatoire des sciences de la Terre de Strasbourg, 1982), GE (<https://doi.org/10.14470/TR560404>) (GEOFON Data Centre, 1993), GO (<https://doi.org/10.7914/1yx9-8844>) (Ilia State University Seismic Monitoring Centre of Georgia, 1988), HE (<https://doi.org/10.14470/UR044600>) (Institute of Seismology, University of Helsinki, 1980), IC (<https://doi.org/10.7914/SN/IC>) (Albuquerque Seismological Laboratory, USGS, 1992), II (<https://doi.org/10.7914/SN/II>) (EarthScope Consortium, 1986), IN (<https://www.fdsn.org/networks/detail/IN/>), IU (<https://doi.org/10.7914/SN/IU>) (Albuquerque Seismological Laboratory, USGS, 1988), KC (<https://doi.org/10.7914/SN/KC>) (Central Asian Institute for Applied Geosciences, 2008), KN (<https://doi.org/10.7914/SN/KN>) (Kyrgyz Institute of Seismology, IVTAN/KIS and University of California, San Diego, 1991), KR (<https://doi.org/10.7914/SN/KR>) (Kyrgyz Institute of Seismology, KIS, 2007), KW (<https://www.fdsn.org/networks/detail/KW/>), KZ (<https://doi.org/10.7914/SN/KZ>) (KNDC/Institute of Geophysical Research (Kazakhstan), 1994), PL (<https://doi.org/10.7914/90rh-0q80>) (Institute of Geophysics, Polish Academy of Sciences, 1990), QZ (<https://doi.org/10.7914/SN/QZ>) (LTD Seismological Experience and Methodology Expedition of the Committee of Science of the Ministry of Education and Science of the Republic of Kazakhstan, 2003), RO (<https://doi.org/10.7914/SN/RO>) (National Institute for Earth Physics, 1994), TJ (<https://doi.org/10.7914/SN/TJ>) (Geophysical Survey of the National Academy of sciences of Tajikistan, 2009), TM (Pornsopin et al., 2023), TU (<https://doi.org/10.7914/SN/TU>) (Disaster and Emergency Management Authority, 1990), XP (https://doi.org/10.7914/SN/XP_2005) (Roecker, 2005), XW (https://doi.org/10.7914/SN/XW_1997) (Roecker et al., 1997), YT (<https://doi.org/10.15778/RESIF.YT2001>) (Poupinet et al., 2001). More details on the seismic networks used in this study are provided in Table S1 in Supporting Information S1. **Software Availability:** The open-access codes used in this study are available as follows: NoisePy for ambient noise preprocessing and cross correlation (Jiang and Denolle, 2020); ThreeStation for three-station interferometry (Zhang et al., 2020); TSAnalysis for teleseismic two-station image analysis (Yao et al., 2005); AFTAN for automatic frequency–time analysis (Bensen et al., 2007); DisperPy for regional earthquake group velocity picking (Nascimento et al., 2025); and SurfATT for adjoint-state surface wave traveltime tomography (Xu et al., 2025).

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